

Table 1: Different GCN and GAT Models Runtime Performance

| Dataset       | GCN Runtime (s) |           | GAT Runtime (s) |           |
|---------------|-----------------|-----------|-----------------|-----------|
|               | DGL(GCN)        | SPLAC-GNN | DGL(GAT)        | SPLAC-GNN |
| pubmed        | 1.92            | 2.94      | 2.45            | 3.52      |
| ogbn-arxiv    | 70.39           | 56.00     | 166.91          | 126.86    |
| ogbn-products | 140.88          | 98.98     | 357.43          | 221.25    |
| reddit        | 150.16          | 59.07     | 280.99          | 109.84    |
| igb-small     | 712.74          | 291.77    | 1168.91         | 467.47    |
| flickr        | 42.03           | 23.93     | 80.48           | 42.84     |